MILITARY SPECIFICATION SHEET

FILTER HOUSINGS, HYDRAULIC, THREE ELEMENTS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-24402.

CLASSIFICATION. Filter housings covered by this specification sheet shall be designated using part or identifying numbers (PIN) as follows:

M	24402	/3	N	B	-WXYZ
Prefix to indicate an item identified by inch-pound units	Specification number	Applicable specification sheet	Type	Element size	Indicator form

Examples: M24402/3NB-G150 is a three element housing without bypass relief using size B elements and provided with a gauge-type differential pressure indicator with a minimum range of 150 lb/in^2 differential.

M24402/3RB is a three element housing with bypass relief using size B elements and provided without a differential pressure indicator.

FILTER ASSEMBLY TYPE. The filter assembly type shall be designated by one of the following letters to indicate whether or not a bypass relief is installed:

N - No bypass relief
R - Bypass relief installed

ELEMENT SIZE. The element configuration and size (see the applicable specification sheet) shall be designated by the following:

B - Size B

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DIFFERENTIAL PRESSURE INDICATOR FORM. This dash number suffix is included in the part number when the filter assembly is to be acquired with a specific type of differential pressure indicator. See the applicable specification sheet for indicator dash numbers. For logistic support purposes, assemblies will be acquired and stocked without differential pressure indicators. In such cases, indicators will be ordered separately using the part number shown on the specification sheet for indicators.

PART NUMBER CROSS REFERENCE. Table I provides a cross-reference between old (MIL-F-24402D and earlier) and new part numbers.

TABLE I. Cross reference of part numbers.

New part number 1/	Old part numbers 2/
M24402/3NB M24402/3RB	M24402-3N-BF-0 M24402-3R-BF-0

1/ If a differential pressure indicator is to be acquired with the filter assembly, add the appropriate indicator dash number suffix from the applicable specification sheet.
2/ Add E, G, or M depending on form of the differential pressure indicator acquired with the filter assembly, as follows:

E - Electrical with mechanical pop-up indicator
G - Gauge type indicator
M - Mechanical pop-up indicator

SUBSTITUTION OF ASSEMBLIES. Assemblies with corresponding old and new part numbers may be used interchangeably provided the correct differential pressure indicator is used.

REQUIREMENTS

(a) Configuration. The filter assemblies shall meet the dimensional requirements shown on figure 1.

(1) Construction requirements. Filter assemblies shall be constructed in accordance with UG of the ASME Boiler and Pressure Vessel Code, Section VIII. Failure of any one bolt or locking element in the head closure shall not result in the failure of other elements and the release of the closure or parts of the closure. The closure shall permit visual external observation to ensure that the holding elements are in good condition and are in full engagement when the closure is in the closed position.

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(b) Maximum rated operating pressure. The filter assemblies shall have a maximum rated operating pressure of 400 lb/in^2.
(c) Rated flow capacity. The filter assemblies shall have a rated flow capacity of 150 gallons per minute.
(d) Performance.

(1) Bypass relief requirements. Type R assemblies shall have bypass relief valve settings as specified in table II.

TABLE II. Bypass relief valve settings.

Test condition	Pressure (lb/in^2)	Test fluid temperature	Leakage rate (mL/min)	
			MIL-L-17331	MIL-H-5606
(a) Cracking pressure				
(1) Qualification	150 minimum	40°C maximum	10 minimum	-
(2) Quality conformance	Not more than 5 lb/in^2 higher than the actual cracking pressure for the qualification assembly.	20 to 60°C	10 minimum	50 minimum
(b) Full flow (qualification only)	200 maximum	40°C maximum	N/A	-
(c) Reseat pressure				
(1) Qualification	120 minimum	40°C minimum	6 maximum	-
(2) Quality conformance	120 minimum	20 to 60°C	6 maximum	12 maximum

(2) Fatigue impulse rating. The fatigue impulse requirement and test of the general specification do not apply.
(3) Proof pressure. The filter assemblies shall withstand a proof pressure of 900 lb/in^2.

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(4) Housing pressure drop. At the maximum rated flow with MIL-L-17331 fluid, the housing pressure drop with minimum restriction dummy elements installed shall not exceed 35 lb/in^2 differential.

(5) Burst pressure. The filter assemblies shall withstand a burst pressure of 1600 lb/in^2.

Preparing activity:
Navy - SH
(Project 4330-N057-03)

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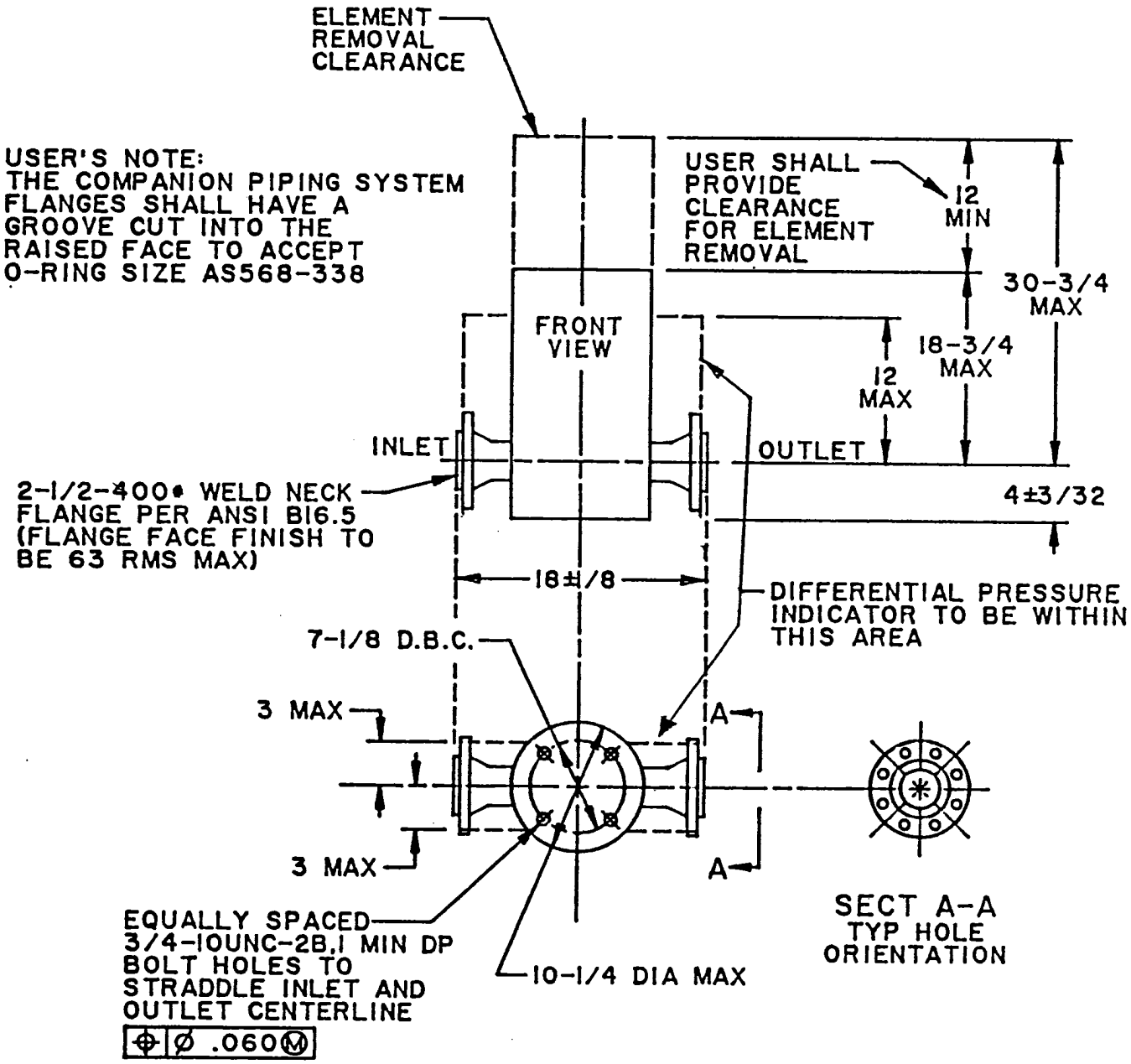


FIGURE 1. Three element filter housing outline dimensions.

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